TEXORA · LECTURE SHEET

Estimation of Pretreated Textile Materials

How the preparation department proves its work: the standard tests for desized, scoured, bleached and mercerised goods — absorbency, residual starch, wax content, whiteness, copper number, fluidity and barium activity number — the faults each preparation step can leave behind, what bad preparation does to dyeing, printing and finishing, and the case for combined and enzymatic processes.

TE-2205 · TEXTILE CHEMISTRY & COLORATION-I · TOPIC 6 · LEVEL 2 TERM 2

1. Why estimation is part of preparation

Preparation faults are invisible until dyeing reveals them — at which point the whole batch is at risk. Estimation (assessment/testing) therefore checks each stage against a measurable property before the goods move on: residual size after desizing, absorbency and residual wax after scouring, whiteness and damage after bleaching, degree of swelling after mercerising. The tests divide into **efficiency tests** (was the impurity removed?) and **damage tests** (was the fibre attacked?). A good preparation passes both.

2. Assessment of desized material

- **Iodine test for starch:** a drop of iodine/potassium-iodide solution on the fabric — blue-black colour means residual starch; its depth is compared against the **TEGEWA scale** (violet-blue stain ratings, 1–9; higher = cleaner) used for rating desizing efficiency.
- **Tests for synthetic sizes:** PVA gives a green-blue colour with iodine–boric acid reagent; CMC and polyacrylates have their own spot reagents — the test must match the size used.
- **Weight-loss check:** percentage loss on desizing + washing should approach the known size add-on; a shortfall signals incomplete removal.

3. Assessment of scoured material

- **Absorbency tests** (the master scouring check): in the **drop test**, a water drop placed on the fabric should disappear within about 1–3 s on well-scoured cotton (minutes on grey fabric); in the **wicking/strip test**, a suspended strip dips into coloured water and the height risen in a fixed time (e.g. 30 min) is measured; in the **sinking test**, a small fabric piece must sink within a few seconds.

- **Residual wax:** solvent extraction (e.g. with carbon tetrachloride/chloroform in a Soxhlet) — grey cotton carries ≈ 0.5–1% wax; well-scoured fabric should retain only ≈ 0.1–0.3% (complete removal is neither possible nor needed).
- **Ash and pectin residues:** residual mineral (ash) and pectic matter (ruthenium-red staining or carbazole test) confirm the scouring chemistry reached the whole load.
- **Evenness:** absorbency is checked at several places across width and length — preparation must be uniform, not merely sufficient on average.

4. Assessment of bleached material

- **Whiteness:** measured instrumentally by reflectance spectrophotometry and reported as a **whiteness index** (CIE or Berger). Guide values: good peroxide full white ≈ 70–85 CIE whiteness (before optical brightener); fluorescent whitening agents push apparent whiteness far higher under UV-containing light.
- **Damage — copper number:** the grams of copper reduced from Cu^{2+} (Fehling-type reagent) per 100 g cellulose. It counts reducing end-groups and oxycellulose sites; it rises with oxidative damage. Well-bleached cotton keeps a low copper number (order 0.1–0.5, typical guide), harshly over-bleached fabric much more.
- **Damage — fluidity:** the fabric is dissolved in cuprammonium hydroxide and the solution's fluidity (reciprocal viscosity, in rhes) measured. Chain cleavage lowers the degree of polymerisation, so fluidity rises with damage: scoured cotton ≈ 2–5 rhes; values climbing well above this after bleaching warn of tendering.
- **Methylene blue absorption:** uptake of the basic dye methylene blue indicates carboxyl groups from oxycellulose — a quick damage indicator.
- **Residual chlorine test** (after hypochlorite bleaching): potassium-iodide/starch paper turns blue if active chlorine remains — it must be fully removed (anti-chlor) or it tenders and yellows goods in storage.

WORKED EXAMPLE — READING A BLEACH REPORT

A peroxide-bleached lot reports: whiteness index 78 (good), absorbency drop test 1.5 s (good), fluidity 9 rhes (high; scoured control was 4), copper number elevated. Diagnosis: the white is excellent but the fibre is oxidatively damaged — the jump in fluidity shows chain cleavage. Before dyeing deep shades this lot risks tendering complaints; the bleach recipe (peroxide dose, temperature, stabiliser, metal control) must be moderated, trading a little whiteness for DP retention.

5. Assessment of mercerised material

- **Barium activity number (BAN):** the ratio of baryta ($Ba(OH)_2$) absorbed by the test fabric to that absorbed by unmercerised cotton, × 100. Unmercerised = 100 by definition; well-mercerised cotton typically reaches ≈ 130–150+; causticised (slack) goods fall between.
- **Iodine absorption / stain tests** and deconvolution microscopy of fibre cross-sections (round vs kidney-bean) support the BAN result.
- **Dye-uptake comparison:** a standard direct-dye test dyeing shows the depth gain mercerisation should have delivered.

6. Faults in the preparation processes

STAGE	TYPICAL FAULT	DOWNSTREAM APPEARANCE
Singeing	Over-singeing (scorch marks); under-singeing (hairiness)	Yellow streaks; fuzzy prints, dye specks, pilling
Desizing	Residual size patches; enzyme killed by wrong pH/temperature	Resist marks — pale, unlevel dyeing exactly where size remains
Scouring	Uneven scouring; wax redeposition; oil spots	Patchy absorbency → cloudy dyeing; spots print as defects
Bleaching	Over-bleaching (oxycellulose); pinholes from metal spots; residual chlorine	Tender fabric, holes, yellowing in storage, unlevel dye uptake (damaged sites dye differently)
Mercerising	Uneven alkali/tension; incomplete washing-out	Lustre variation, streaky dye depth across the width
All stages	Crease marks set in rope processing; water spots; mildew on stored wet goods	Permanent lines and stains visible after dyeing

7. Impact of improper preparation on dyeing, printing and finishing

Dyeing is an audit of preparation. Residual size or wax acts as a resist, so dye shades come out unlevel, pale in patches and poor in rubbing fastness. Damaged (oxidised) cellulose carries extra carboxyl groups: it attracts basic and direct dyes differently, repels some classes, and loses strength until seams fail in garmenting. Unremoved seed fragments appear as dark specks under light shades. In printing, poor absorbency gives blurred outlines, low colour yield and print-paste marks that will not wash off; in finishing, residues interfere with resin and softener uniformity, producing patchy easy-care performance

and handle. The economic conclusion follows: re-preparing and re-dyeing a faulty batch costs far more than the tests in Sections 2–5 that would have caught the fault in time.

8. Combined preparatory processes and energy conservation

Steps are increasingly **combined**: one-bath desize–scour, and scour–bleach, in pad-steam continuous ranges; cold pad-batch combinations; and "one-step" preparation of knits where scouring and bleaching share one bath. Combining saves water (fewer baths and rinses), energy (fewer heat-ups) and time — pad-steam lines treat fabric in minutes rather than the kier's hours. Energy conservation around them includes counter-current washing, heat recovery from hot effluent, low-liquor machines and full loads. The trade-off is control: combined baths demand precise dosing (alkali, peroxide, stabiliser, enzyme compatibility) because each step can no longer be adjusted independently.

9. Benefits of enzymatic preparatory processes

- **Specificity:** amylase attacks only starch, pectinase only pectin, catalase only residual peroxide — no side attack on fibre when dosed correctly.
- **Milder conditions:** 50–70 °C, near-neutral pH → less energy, less cellulose damage, softer fabric handle, lower weight loss.
- **Effluent quality:** enzymatic desize effluent has lower chemical load than acid/oxidative desizing; bio-scouring cuts alkali and TDS discharge.
- **Safety and handling:** no corrosive bulk chemicals for those steps.
- Limitations to manage: enzymes are inactivated by extremes of pH/temperature and by some surfactants and metal ions, cost more per bath, and bio-scouring removes wax less completely than caustic scouring — heavy-duty whites may still need an alkaline stage.

10. Glossary — key terms

- **Copper number:** grams of Cu reduced per 100 g cellulose; a measure of oxidative damage.
- **Fluidity:** reciprocal viscosity of a cuprammonium solution of the cellulose (rhes); rises as the polymer chain is cut.
- **Barium activity number:** relative $Ba(OH)_2$ absorption measuring degree of mercerisation (unmercerised = 100).
- **TEGEWA scale:** standard stained-scale (1–9) for rating residual starch after desizing.
- **Whiteness index:** instrumental number (CIE/Berger) expressing reflectance-based whiteness.

11. Key points

- Test each stage: iodine/TEGEWA for size, absorbency and wax for scouring, whiteness plus copper number/fluidity for bleaching, BAN for mercerising.
- Efficiency and damage must be checked together — a very white fabric may be a tender fabric.
- Every preparation fault reappears downstream as a dyeing, printing or finishing fault.
- Combined processes and low-liquor, heat-recovered equipment save water and energy; enzymes add specificity and mildness at higher chemical cost.

12. Summary

Estimation turns preparation from habit into control. Spot and instrumental tests quantify what each step achieved — absorbency for scouring, whiteness for bleaching — and, just as importantly, what it cost the fibre, through copper number and fluidity. The fault table shows why this matters: dyeing, printing and finishing simply expose preparation's mistakes. Modern preparation answers with combined processes, energy recovery and enzyme technology, all verified by the same battery of tests.

13. Practice

Answer in your own words; brief model answers follow.

1. A desized fabric gives a strong blue colour with iodine solution and a TEGEWA rating of 2. Interpret the result and state the corrective action.
2. Why are copper number and fluidity both called damage tests, and what molecular event do they each detect?
3. Two bleached lots: A — whiteness 80, fluidity 4 rhes; B — whiteness 84, fluidity 11 rhes. Which lot do you accept for a high-quality shirting, and why?
4. Give three preparation faults and the exact dyeing/printing defect each produces.
5. A mill replaces acid desizing and caustic scouring with amylase desizing and pectinase bio-scouring. List two benefits and two limitations it should expect.

Answer key

1. Heavy residual starch remains (rating 2 is poor; high ratings approach complete removal). The fabric must not proceed: re-desize (check enzyme activity, pH, temperature and dwell), wash thoroughly, and re-test before scouring.
2. Both detect cellulose chain degradation. Copper number counts reducing groups (new chain ends and oxidised sites) created by cleavage/oxidation; fluidity measures the resulting fall in degree of polymerisation through solution viscosity. They rise together when fibre is damaged.

3. Lot A. Its fluidity shows the cellulose DP is essentially intact, while B's high fluidity signals serious oxidative tendering; 4 whiteness points are not worth a fabric that may fail in wear and garmenting.
4. Any three, e.g.: residual size → resist marks/pale patches in dyeing; over-bleaching (oxycellulose) → unlevel dye uptake and tendering; uneven scouring → cloudy, patchy dyeing; residual chlorine → yellowing and tendering in storage; under-singeing → fuzzy print outlines and specks.
5. Benefits (any two): lower temperature and near-neutral pH save energy and reduce fibre damage/weight loss; milder effluent (lower TDS, no acid); specific action with softer handle. Limitations (any two): higher chemical cost; enzymes are inactivated by wrong pH/temperature/metal ions; bio-scouring leaves more residual wax than caustic scouring, which can limit top-whiteness work.

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